Marking Out Tools for Wood, Metal and Acrylic

Marking out is the process of transferring a design or pattern to a workpiece, as a guide for cutting, drilling or machining. Marking out tools are used to measure, mark and check the accuracy of the workpiece. Different materials may require different tools for marking out, depending on their properties and characteristics.

In this worksheet, you will learn about some common marking out tools used for wood, metal and acrylic, and answer some questions to test your understanding.

Tape Measure

A tape measure is a flexible ruler that can measure straight and curved shapes. It is usually made of metal or plastic, and has markings in metric or imperial units. Tape measures are used to measure the length, width and thickness of workpieces, as well as the distance between points or features.

Questions:

1. What are the advantages and disadvantages of using a tape measure for marking out?
2. How can you ensure accuracy when using a tape measure?
3. What are some common errors or mistakes when using a tape measure?

Pencil

A pencil is a simple tool that can be used to mark lines or points on workpieces. It is usually made of graphite or lead, and has a wooden or plastic casing. Pencils are suitable for marking out on wood and acrylic, but not on metal, as the marks can be easily rubbed off or obscured by cutting fluids.

Questions:

4. What are the advantages and disadvantages of using a pencil for marking out?
5. How can you ensure accuracy when using a pencil?
6. What are some common errors or mistakes when using a pencil?

Craft Knife

A craft knife is a sharp tool that can be used to cut and score (mark) workpieces, particularly on paper and card. It is usually made of metal, and has a replaceable blade. Craft knives are suitable for marking out on wood and acrylic, but not on metal, as the blade can be damaged or dulled by the hard surface.

Questions:

7. What are the advantages and disadvantages of using a craft knife for marking out?
8. How can you ensure accuracy when using a craft knife?
9. What are some common errors or mistakes when using a craft knife?

Compass

A compass is a tool that can be used to mark out circles or arcs on workpieces. It is usually made of metal, and has two legs that can be adjusted to set the radius of the circle. One leg has a sharp point that acts as the centre of the circle, and the other leg has a pencil or a scribe that marks the circumference of the circle.

Questions:

10. What are the advantages and disadvantages of using a compass for marking out?
11. How can you ensure accuracy when using a compass?
12. What are some common errors or mistakes when using a compass?

Try Square

A try square is a tool that can be used to mark out angles that are 90 degrees to an edge. It is usually made of metal or wood, and has a stock and a blade that form a right angle. The stock is placed against the edge of the workpiece, and the blade is used to mark or check the perpendicular line.

Questions:

13. What are the advantages and disadvantages of using a try square for marking out?
14. How can you ensure accuracy when using a try square?
15. What are some common errors or mistakes when using a try square?

Mitre Square

A mitre square is a tool that can be used to mark out angles that are 45 degrees to an edge. It is similar to a try square, but has a blade that forms a 45 degree angle with the stock. The stock is placed against the edge of the workpiece, and the blade is used to mark or check the mitre line.

Questions:

16. What are the advantages and disadvantages of using a mitre square for marking out?
17. How can you ensure accuracy when using a mitre square?
18. What are some common errors or mistakes when using a mitre square?

Scriber

A scriber is a tool that can be used to mark lines or points on metal workpieces. It is usually made of steel, and has a sharp point that can scratch the surface of the metal. Scribers are suitable for marking out on metal, but not on wood or acrylic, as the marks can be too faint or damage the material.

Questions:

19. What are the advantages and disadvantages of using a scriber for marking out?
20. How can you ensure accuracy when using a scriber?
21. What are some common errors or mistakes when using a scriber?

Centre Punch

A centre punch is a tool that can be used to mark points on metal workpieces, especially for drilling. It is usually made of steel, and has a tapered point that can create a small indentation on the metal. A hammer is used to strike the centre punch, which transfers the force to the point. Centre punches are suitable for marking out on metal, but not on wood or acrylic, as the marks can be too deep or crack the material.

Questions:

22. What are the advantages and disadvantages of using a centre punch for marking out?

23. How can you ensure accuracy when using a centre punch?

24. What are some common errors or mistakes when using a centre punch?

Combination Square

A combination square is a tool that can be used to mark out various angles and measurements on workpieces. It is usually made of metal, and has a ruler and a head that can slide along the ruler. The head has a 90 degree angle, a 45 degree angle, a spirit level and a scribe. Combination squares are suitable for marking out on wood, metal and acrylic, as they can measure and mark different features.

Questions:

25. What are the advantages and disadvantages of using a combination square for marking out?
26. How can you ensure accuracy when using a combination square?
27. What are some common errors or mistakes when using a combination square?

Marking Gauge

A marking gauge is a tool that can be used to mark parallel lines to an edge on workpieces. It is usually made of wood or metal, and has a stem, a head and a pin or a wheel. The head can be adjusted to set the distance from the edge, and the pin or the wheel can mark the line on the workpiece. Marking gauges are suitable for marking out on wood and acrylic, but not on metal, as the pin or the wheel can be worn out or slip on the surface.

Questions:

28. What are the advantages and disadvantages of using a marking gauge for marking out?

29. How can you ensure accuracy when using a marking gauge?

30. What are some common errors or mistakes when using a marking gauge?